

14 > Fix your Wi-Fi first

Pick cleaner 2.4 GHz channels

In most places, channels 1, 6, and 11 are the only non-overlapping options on 2.4 GHz, which is why they're the standard recommendation for reducing interference.

Step-by-step: set 2.4 GHz channel sanely

1. If your router has "Auto channel," keep it for 5 GHz, but consider manual for 2.4 GHz.
2. Choose 1 or 6 or 11 (try each for a day if needed).
3. Re-test 2–3 problem spots (from your coverage map). Keep the channel that gives fewer dropouts, not just higher speed.

Decide whether you need Smart Connect / band steering

Many routers merge bands under one SSID and automatically push devices to a band. Netgear calls this "Smart Connect," for example, where one SSID covers 2.4 and 5 GHz and the router selects the band per device.

Band steering is great for phones and laptops – but it can confuse IoT onboarding and some picky 2.4-only devices. If you're struggling, temporarily split SSIDs. Netgear explicitly notes you can create separate SSIDs for 2.4 and 5 GHz on some mesh products, and that this helps keep older/slower devices on 2.4 while reserving 5 GHz for high-speed devices.

Step-by-step: the "setup mode" trick

1. Split SSIDs (or create a 2.4-only IoT SSID).
2. On your phone, connect to the 2.4 SSID during device setup.
3. After onboarding, keep the IoT device on the 2.4 SSID permanently.
4. If you want, you can re-enable Smart Connect later for your main SSID – just don't move IoT devices off their stable lane.

5) Mesh Wi-Fi basics

Mesh works when nodes have a strong connection to each other. Topology matters too.

According to various reports, daisy-chaining nodes (tree topology) add speed penalties with each hop, and systems prefer star topology when connection quality allows. Netgear's guidance: mesh signals should slightly overlap, and a rough rule is one node for every two rooms (layout and walls will change this).

A useful mental model from hands-on mesh setup guides: don't place a node "inside" the dead zone – place it where it can still get a strong signal from the main node, then extend coverage outward.

Step-by-step mesh placement walkthrough

1. Place the main node (router) in the best central spot you can manage.
2. Put node #2 at the Green/Amber boundary – not deep in Red.
3. Check backhaul quality in the mesh app; if it's weak, move the node closer.

4. Add node #3 only if needed, and try to place it so it connects directly to the main node (star).
5. Re-run the dead-zone test after every move.

Multi-floor homes: a simple placement heuristic

- Put the main node on the floor where most usage happens (often the living room).
- Place a second node near the staircase/vertical "spine" of the home.
- If possible, use wired backhaul between floors (even one Ethernet run helps).

Wired backhaul: the cheat code

If you can run Ethernet to one node, do it. Wired backhaul improves stability and avoids hop-related slowdowns (especially in long homes and multi-floor layouts).

6) The "IoT lane"

You're doing this because:

- IoT devices are often less secure than laptops/phones
- many devices on one lane can cause discovery/crowding issues
- you want a clean way to reset IoT later without touching your main devices

A U.S. government cybersecurity guide recommends home network segmentation – at minimum separating primary Wi-Fi, guest Wi-Fi, and an IoT network – to keep less secure devices from directly communicating with more secure devices.

Pick one of these levels.

Level 1 (easy): Guest network = IoT lane

Pros: quick isolation, supported on most routers.

Cons: guest isolation can sometimes block local discovery (casting, device pairing quirks).

Google/Nest Wi-Fi (Guest network)

1. Open the Google Home app.
2. Tap Favourites → Wi-Fi → Network settings → Guest network.
3. Turn it on, set name/password, save.
4. Put smart-home devices on this lane.
5. Test what you use: casting, speaker groups, device discovery, automations.

TP-Link routers (Guest network)

1. Log into the router interface/app.
2. Go to Advanced → Guest Network.
3. Enable guest network on 2.4 GHz (and 5 GHz if you want).
4. Set SSID/password and save.

If your router offers permissions like "allow guests to see each other," keep it OFF unless you specifically need IoT devices to talk to each other inside the lane.

Name lanes clearly:

- Home_Main